

Description of Scholarly Experience, Interests and Goals

Presented to the Committee on Appointments and Promotions

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My undergraduate research opportunity at Notre Dame's Radiation Laboratory (results published as "Monomer-eximer dynamics in spread monolayers", 1980¹) fostered in me a desire for a lifetime of scientific research. At the same time, my Notre Dame undergraduate majors in chemistry and philosophy made it clear to me that science makes progress and generates agreement, whereas moral theory does neither. Given the importance of morality in my own life, at Notre Dame and in Catholic intellectual traditions, I found this difference between science and morality both mystifying and unfortunate; I was drawn from science to graduate school in philosophy to try to understand and perhaps rectify this state of affairs. There I found some measure of the understanding I sought in the work of Alasdair MacIntyre, beginning with my reading of his *Short History of Ethics* in 1980 and *After Virtue* in 1981. Following his lead, I set out to gain an adequate preparation for moral theory in the broadly Thomistic tradition by specializing in the metaphysics of Aristotle and the medieval Aristotelian tradition. With support from a Charlotte W. Newcombe Doctoral Dissertation Fellowship (1985-86) I wrote an interpretation of Aquinas as a compatibilist with respect to issues of causation and human free choice, and I understood that interpretation to hold promise for a new articulation of Thomistic

moral theory. (I reworked and published the dissertation material much later as “Aquinas, Compatibilist” in 1999².)

My efforts to articulate a Thomistic moral theory on a compatibilist foundation were colored by my early-career teaching responsibilities, chiefly in applied ethics. (I taught 20 sections of Engineering Ethics over seven years at the University of Portland.) I worked to responsibly present MacIntyre’s account of the present state of moral confusion in a way which did justice to the problem of moral relativism and yet suggested a way forward intellectually and practically: by choosing to inhabit a tradition that upheld respect for human dignity. I worked towards an account of such respect (“Resanctifying Human Life”, 1990³, “What is Respect for the Sanctity of Life?”, 1991⁴) and toward a broadly Thomistic moral theory embodying such respect (“Freedom and Good in the Thomistic Tradition”, 1994⁵, and “Three Rival Versions of Nonmoral Goodness”, 1995⁶.) As these titles indicate, I was still following MacIntyre’s developing thought with interest. His account of modern moral relativism, as well as his exploration of a remedy for our age, rested entirely on his general epistemological account, and particularly on his qualified acceptance of Kuhnian relativism in science and his provisional overcoming of that relativism in holding out the recovery of a tradition of holistic inquiry. As I attempted to develop moral theory in a way shaped by MacIntyre’s work, I became aware of the need to understand how formation in a scientific tradition of inquiry actually takes place, and to address the need for a tradition of holistic inquiry from a perspective of credible familiarity with how science is actually conducted. I could not imagine how a philosopher could acquire such a perspective. I

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turned to high school teaching, immersed myself in the dialogues of Plato, and began to outline a book I had been developing since 1990, a defense of the sanctity of human life in MacIntyre's craft-tradition context. (I presented "The Narrative Context of Moral Theory" to the Summer Thomistic Institute at Notre Dame in 1996⁷ as part of that effort.)

After three years teaching philosophy and literature at the high school level, I was asked to teach high school chemistry (and eventually geometry, precalculus, and physics.) I began to do so in 1997. The next year I learned of an opportunity to participate in Notre Dame's particle physics research community as part of the QuarkNet program. I began meeting weekly with Randy Ruchti and some dozen local high school teachers in Fall 1999, and have continued meeting weekly and working full time during summers at the Notre Dame QuarkNet Center (NDQC) since that time. My early efforts at the NDQC were devoted to understanding the Standard Model and the physics and engineering behind the production of high energy particle accelerators and detectors. I participated in the design of fiber layout in hadronic calorimeter HB optical decoding units (ODU) for the Compact Muon Solenoid (CMS) detector, and ODU module insertion in the readout boxes for the Hadronic Calorimeter (HCAL) ("Electrico-optical Interface Design for CMS HCAL", Marchant, Loughran et al., 2002⁸, and "Electrico-Optical Readout System for the CMS Scintillating Hadronic Calorimeters", Karmard, Loughran et al., 2000⁹.)

I continued to monitor CMS detector development as a sponsor of high school students participating in the construction of detector elements, but my NDQC involvement turned largely

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toward education and public outreach (E/PO) concerns in 2001. I agreed to help work out a collaboration between QuarkNet and the Modeling physics instructional community at Arizona State University, strongly suggested by principals at NSF. With two other local teachers I traveled to ASU for three weeks to get to know their community and the Modeling method, and to plan a joint national conference with them for QuarkNet teachers. I helped design and coordinated the QuarkNet Modeling Workshop, a three week collaboration between these two communities held at Notre Dame in summer 2002. I have since twice used the Modeling method as a high school physics instructor, and have read the Modeling listserve on nearly a daily basis for more than four years. The Modelers represent a unique pedagogical community I intend to continue to monitor.

During those same years I developed an E/PO affiliation with a nanotechnology initiative at Notre Dame. Several faculty from Notre Dame's Electrical Engineering and Chemistry departments achieved finalist status on a \$10M NSF Nanotechnology Center grant proposal (DOMINO); I had agreed to help design an outreach to middle and high school students as part of that proposal, and suggested that we replicate around nanoelectronics the kind of open-ended invitation to teachers that was beginning to flourish around high energy physics at NDQC. I presented that idea to NSF in a reverse site visit evaluation of the DOMINO proposal. That proposal did not ultimately succeed. But articulating the model that had been working for two years at the NDQC became an occasion for understanding better how scientific research could become the focal point for the professionalization of precollege science teaching, which my experience with the ASU modeling program had convinced me could become a craft, though it plainly has not yet risen to that level on any large scale. So I began to articulate the model,

which I called the Research Community for Teachers model, after the NSF's Research Experience for Teachers (RET) program which has been supporting our efforts at NDQC. I presented that model at several national professional meetings. I also continued my collaboration with the nanoelectronics community, and currently have charge of a Scanning Tunneling Microscope for E/PO purposes as a Postdoctoral Research Associate in Electrical Engineering.

The move toward professionalization of pre-college science teaching began to take hold at the NDQC. We began aiming to develop expertise in providing student research opportunities. Toward this end I applied for and became a member of the 2004 Teacher Leaders in Research Based Science Education (TLRBSE) program of the National Optical Astronomy Observatory, so that we might learn from their experience of providing research opportunities for teachers and students. Shortly thereafter I moved to Saint Joseph's High School from Trinity School, and established the Saint Joseph's High School Research Community (SJHSRC). My aim was to promote student contributions to science and science education, in part through student research projects, though decidedly not on an individual basis, but rather with explicit attention to the collaborative nature of science. During that time I served as the leader of student research at the NDQC. I applied for and became a 2006 Spitzer Research Teacher, which together with my connections with TLRBSE, QuarkNet and the ND nanoelectronics group formed a constellation of science research opportunities for students in the SJHSRC to help develop.

One additional strand of these research opportunities for teachers and students grew to prominence during these years. Ever since the 2002 QuarkNet Modeling Workshop we had

participated in a growing discussion at the national QuarkNet level concerning the advent of the Large Hadron Collider (LHC) and two of its detectors, CMS and Atlas. These experiments were due to come on line in late 2007, and from its inception QuarkNet was designed to prepare high school teachers and (indirectly) students to appreciate and even participate in LHC research. The national QuarkNet collaboration eventually turned to participation in cosmic ray research, since the detectors are relatively inexpensive to build, and the accelerator is free. At the same time, preparations for the high performance computing dimensions of LHC research had led toward the development of Grid computing tools; these cosmic ray and Grid emphases have been joined together in the idea of an e-Lab.

e-Labs are Grid-enabled online research environments which enable students to efficiently climb the learning curve (meeting a series of performance milestones linked to references) into investigation of data using the data analysis tools of large-scale scientific collaborations.

Beginning with the cosmic ray research collaborative which was launched in 2002 at NDQC in cooperation with the national QuarkNet office, this effort has now become funded through the NSF-funded Interactions in Understanding the Universe (I2U2) program, administered out of the Education Program Office at Fermilab. In January 2006 I was hired as Education Program Leader for I2U2. Currently I work not only with the existing cosmic ray collaboration, but in developing e-Labs for CMS, the Laser Interferometer Gravitational-Wave Observatory (LIGO) at Hanford, the STAR detector for the Relativistic Heavy Ion Collider at Brookhaven National Lab, and the Mariachi cosmic ray detection project. I am also co-PI on the NDeRC GK-12 proposal to involve graduate students in e-Lab development and implementation in area schools.

I plan to continue to develop both the RCT model of professional development of pre-college science educators, as well as the student research emphasis of the research community for teachers at NDQC, as I continue my work for I2U2 developing research environments for high school science students. Having returned to science from my quest to understand the foundations of moral theory, I intend to better understand how science progresses, and to find ways to develop a craft of science education by hitching the professional development of science teachers to the engine of scientific research. The success or failure of that endeavor might shed light on the prospects of developing the craft of living well in the context of a tradition of holistic inquiry. I believe this is the path forward toward development of the sort of community which might claim to make moral progress and overcome moral relativism in the way that the sciences have progressed and generated consensus.

The text above, written to ND's Committee on Appointments and Promotions, is left as written with the exception of the inclusion of footnotes. Below, I pick up that narrative, some 18 years later.

- Design, wrote and managed NDeRC (NSF xxxxxx \$2.7M was funded (ISTEM community model) 2006? (a few public highlights); I related to NSF for annual PI meetings from 2007-2013
- Renewal application to state of IN in 2013? with 1M in ND matching funds committed (who, how much); unsuccessful but a sign of widespread ND impact
- Left I2U2 during funding crisis

- ISTEM Community conferences (Collaborating for STEM Research, Education and Commercialization Forum I - XIII, 2007 - 2020) (highlights from last summary sheet, including remote in 2020)
- BOSCO 2006; Breaking Borders Award 2009?;
- Accenture grants CE3 2012-? \$2.7M; Through 2017? (BOTFL trip)
- Advised SIBC from x to y; SIBC trip)
- enFocus 2013 - 2024
- Work with molecular technology crowd continued from x to y (pull from NISE application?)
- Continued work with Spitzer, SJHSRC (as pro-bono instructor of research class); STAR formation rate paper published xxx; NITARP
- Service as NSF reviewer began with GK-12 in 2006; continued over 8 programs in 20 rounds of review
- Always soft-money SPF, appointed full PF in 2006; P of Practice in XXXX;
- INTEGRITY in xxxx; xxxxy with ND Energy; 100 campus interviews; IGE version competed but turned away for want of matching funds
- Moved to part-time adjunct when INTEGRITY did not succeed
- Tried a couple of start-ups (a new engine design with non-ND personnel; an e coli test with undergrads working with professors at Ivy Tech and eventually Marya Liberman and Holly xxxx in ND chemistry, serving as acting CEO for K&L Biotech (writing successful patent appeal for xxxxxxxx with friends/family funding)
- Began Resurrecting Civility and a companion anthology (Toward Computational Natural Law) in 2017 - current

- Recruited partners (eventually, Prof.s Wendy and eventually Corey Angst) for the rebuilding of St. Bakhita's a girls vocational school in Kalongo, Northern Uganda (at the request of the Archbishop of Gulu, the Most Reverend John Baptist Odama.) Spent 1 year in Prof. Wendy Angst's Design Thinking course, helping to launch multiple projects and organizing and guiding a Spring Break trip for 14 students and Prof. Angst just prior to the COVID shut-down of the University. (This effort was later featured in an NBC *What Would You Fight For* spot.)
- Accepted invitation to teach Physics A (Mechanics) and B (E & M) labs to freshman Physics in AY 2020-21 and 21-22.
- Spun out and presented two papers: Toward a Computational Natural Law Defense of the Sanctity of Human Life to the DeNicola Center for Ethics and Culture Fall Conference in Fall 2022; Four Models of Cultural Exchange: Aesthetic, Moral, Prescientific and Sapiential to the International Society for MacIntyrean Inquiry in Summer 2023.
- Accepted Instructional Resource Program Manager position starting April 2022. Created & published a running inventory of availability of 241 physics and astronomy demos (currently at 96.7% ready.)
- Biogas effort 2022-current (financial model built for providing biogas to rural households across all of Northern Uganda);